

2: Tooling & Execution Setup (JMeter, Taurus, Jenkins)

Purpose

This page documents the technical setup and execution parameters for IDP login performance testing. It serves as a reference for teams running tests locally, via Jenkins, or through BlazeMeter.

1. Tooling Overview

- **JMeter**
 - Core scripting tool (JMX files).
 - Most scripts have YAML wrappers for Taurus.
 - Auth, IDP login, account, metadata flows already scripted.
- **Taurus**
 - Orchestrates JMeter runs with YAML configs.
 - Enables parameterization (concurrency, duration, ramp).
 - Direct integration to BlazeMeter for centralized reporting.
- **Jenkins (AGSUP_ENDURANCE)**
 - Central CI/CD runner for endurance and baseline runs.
 - Accessible at: [Jenkins Job – AGSUP_ENDURANCE](#)
 - Accepts parameters (server, YAML file, environment, concurrency, duration, ramp).
- **BlazeMeter**
 - Repository of historical reports.
 - Supports baseline comparisons across dates.

2. Standard Parameters

When triggering from Jenkins:

- **server:** loadtestwt1 / loadtestwt2 (Use WIN SCP - Credentials are VDI/RSD/Windows login one)
- **yaml:** Path to Taurus YAML (e.g., idp/auth-server-remote.yaml, idp/idp-login-resources-remote.yaml)
- **environment:** Typically stage1 (Optional QC4)
- **encrypted:** Credential flag (always checked)
- **concurrency:** Number of virtual users (e.g., 1, 10, 25, 50)
- **duration:** Execution length (e.g., 1m, 10m, 1h)
- **ramp:** Ramp-up time (e.g., 1s, 30s)

Examples (from recent jobs):

- **10 users / 10m run / 30s ramp** – baseline auth server validation.
- **25 users / 1h run / 30s ramp** – endurance IDP login resources flow.
- **1 user / 1m run / 1s ramp** – quick smoke validation.

3. BlazeMeter Reporting

Centralized reporting dashboard for analysis and leadership visibility.

Access Reports:

- BlazeMeter automatically captures runs executed via Taurus/Jenkins.
- Example: [Sample BlazeMeter Report Link](#).
- Example: [BlazeMeter Report Dashboard](#).
- **Report Contents:**
 - Avg throughput, response time (avg/p90/p95).
 - Error % (HTTP + assertions).
 - Timeline charts for VU load vs response.
 - Assertions for login/account overview validation.
- **Example Snapshot (from recent run):**
 - 25 VU, 57m run.
 - Avg Throughput: **50.08 hits/s**.
 - Avg Response: **404ms**.
 - 90% Response: **1142ms**.
 - Errors: **0.06%**.

4. Existing Project Coverage

As per Nick's [documentation](#):

- **IDP Login Flows:**
 - `idp-login` (PKCE CS + API calls, requires MFA disabled).
 - `idp-login-resources` (PKCE + static resource calls, includes MFA traffic only).
- **Auth Server Flows:**
 - `auth-server` (Client ID/Secret, token introspect, userinfo).
 - `auth-server-delay` (same with random execution delay).
- **Account Services:**
 - Account enrollment, metadata fetch, withdrawals.
 - Most API-only flows, CSV-driven test data.
- **UI Flows (via Taurus Selenium/Apirtif):**
 - Local + Selenium Grid versions.
 - All configured for BlazeMeter reporting.

5. Best Practices for Running Tests

1. **Start Small Scale Up**
 - Run a **smoke test** (1–2 users, 1m) before large loads.
 - Catch immediate environment/data issues.
2. **Use Consistent Parameters**
 - Always note concurrency, duration, ramp time in Confluence for traceability.
3. **Keep Data Ready**
 - Ensure CSVs for accounts, credentials, UUIDs are up to date.
 - MFA should be **disabled** for login test accounts.
4. **Baseline Before Endurance**
 - Validate flows at 10–25 users before running 8–10 hr tests.
5. **Use BlazeMeter as Source of Truth**
 - Upload all runs for centralized visibility.
 - Tag reports by flow and date.
6. **Automate Where Possible**
 - Configure Jenkins to trigger nightly runs.
 - Automate pass/fail validation against SLA thresholds (from Child Page 1).

6. Troubleshooting Tips

- **High Errors?** Check CSV data, credentials, or environment availability.
- **Slow Response Times?** Validate DB locks or infra bottlenecks.
- **BlazeMeter Upload Missing?** Re-run with `-report` enabled in YAML.
- **MFA failures?** Ensure MFA disabled for test accounts.

5. Quick Execution Guide

1. Go to [Jenkins Job – AGSUP_ENDURANCE](#).
2. Select **Build with Parameters**.
3. Enter:
 - **server** (loadtestwt1 / loadtestwt2).
 - **yaml** (choose correct YAML path).
 - **environment** = stage1.
 - **concurrency, duration, ramp** as per test plan.
4. Run job.
5. After execution:
 - Verify BlazeMeter report link.
 - Record results under "BlazeMeter Baseline" Confluence page.